CHAPTER 7:

# Clustering



# Clustering

- Unsupervised classification

- Some Important Applications
  - Categorization
  - Visualization
  - Preprocessing

# Clustering: Types

- Exclusive
  - One object belongs to exactly one cluster
- Overlapping
  - An object may belong to more than one cluster
- Fuzzy
  - Membership weights, between 0 and 1.

# Classes vs. Clusters

- **Supervised**: $X = \{ \boldsymbol{x}^t, \boldsymbol{r}^t \}_t$
- Classes $C_i$ $i=1,\ldots,K$

$$p(\mathbf{x}) = \sum_{i=1}^{K} p(\mathbf{x} \mid C_i) P(C_i)$$

where $p(\boldsymbol{x} \mid C_i) \sim N(\boldsymbol{\mu}_i, \boldsymbol{\sum}_i)$

- $\Phi = \{P(C_i), \boldsymbol{\mu}_i, \boldsymbol{\sum}_i\}^K_{i=1}$

$$\hat{P}(C_i) = \frac{\sum_t r_i^t}{N} \quad \mathbf{m}_i = \frac{\sum_t r_i^t \mathbf{x}^t}{\sum_t r_i^t}$$

$$\mathbf{S}_i = \frac{\sum_t r_i^t (\mathbf{x}^t - \mathbf{m}_i)(\mathbf{x}^t - \mathbf{m}_i)^T}{\sum_t r_i^t}$$

- **Unsupervised** : $X = \{ \boldsymbol{x}^t \}_t$
- Clusters $G_i$ $i=1,\ldots,k$

$$p(\mathbf{x}) = \sum_{i=1}^{k} p(\mathbf{x} \mid G_i) P(G_i)$$

where $p(\boldsymbol{x} \mid G_i) \sim N(\boldsymbol{\mu}_i, \boldsymbol{\sum}_i)$

- $\Phi = \{P(G_i), \boldsymbol{\mu}_i, \boldsymbol{\sum}_i\}^k_{i=1}$

Labels, $\boldsymbol{r}^t_i$ ?

# *k*-Means Clustering

- Partition clustering: non-overlapping clusters
- A variation: K-medoids (use median instead of mean)
- Find *k* reference vectors (prototypes/codebook vectors/codewords) which best represent data
- Reference vectors, $\boldsymbol{m}_j$, $j = 1,\ldots,k$
- Use nearest (most similar) reference:

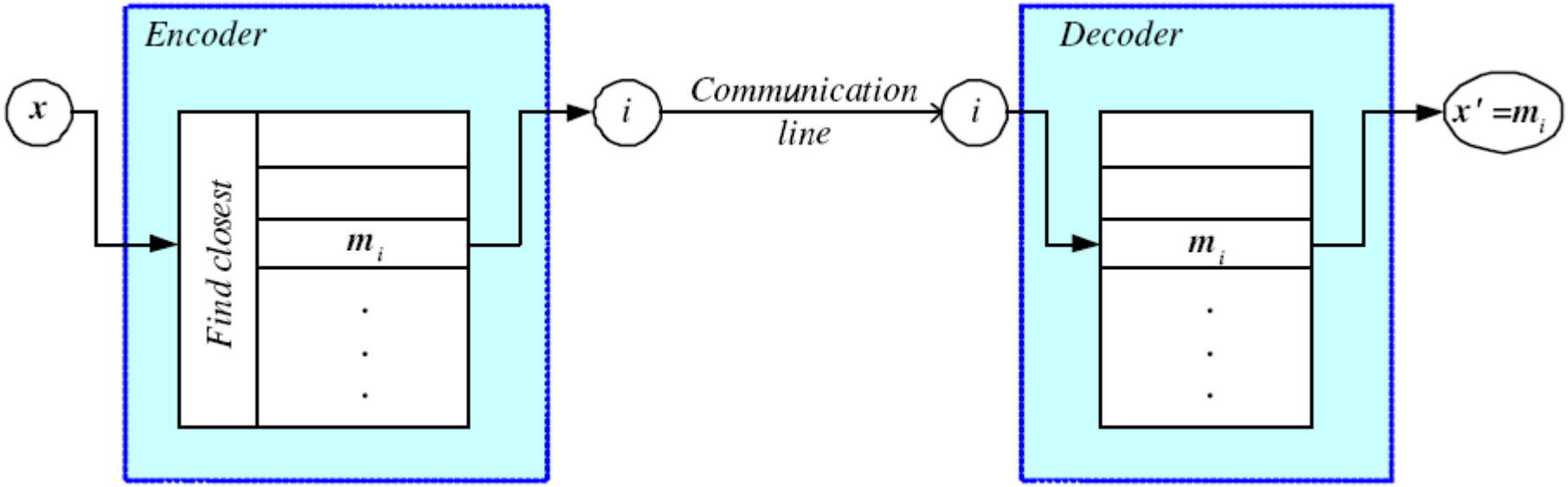
$$\left\|\mathbf{x}^t - \mathbf{m}_i\right\| = \min_j \left\|\mathbf{x}^t - \mathbf{m}_j\right\|$$

- Reconstruction error

$$E\left(\{\mathbf{m}_i\}_{i=1}^k | \mathcal{X}\right) = \sum_t \sum_i b_i^t \left\|\mathbf{x}^t - \mathbf{m}_i\right\|$$

$$b_i^t = \begin{cases} 1 & \text{if } \left\|\mathbf{x}^t - \mathbf{m}_i\right\| = \min_j \left\|\mathbf{x}^t - \mathbf{m}_j\right\| \\ 0 & \text{otherwise} \end{cases}$$

# Encoding/Decoding



# k-means Clustering Algorithm

Initialize $\boldsymbol{m}_i, i = 1, \ldots, k$, for example, to $k$ random $\boldsymbol{x}^t$

Repeat

For all $\boldsymbol{x}^t \in \mathcal{X}$

$$b_i^t \leftarrow \begin{cases} 1 & \text{if } \|\boldsymbol{x}^t - \boldsymbol{m}_i\| = \min_j \|\boldsymbol{x}^t - \boldsymbol{m}_j\| \\ 0 & \text{otherwise} \end{cases}$$

For all $\boldsymbol{m}_i, i = 1, \ldots, k$

$$\boldsymbol{m}_i \leftarrow \sum_t b_i^t \boldsymbol{x}^t / \sum_t b_i^t$$

Until $\boldsymbol{m}_i$ converge

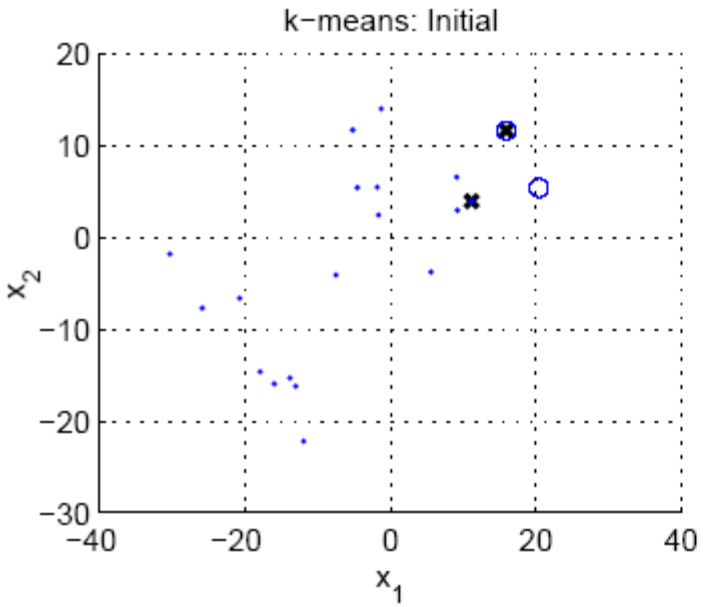


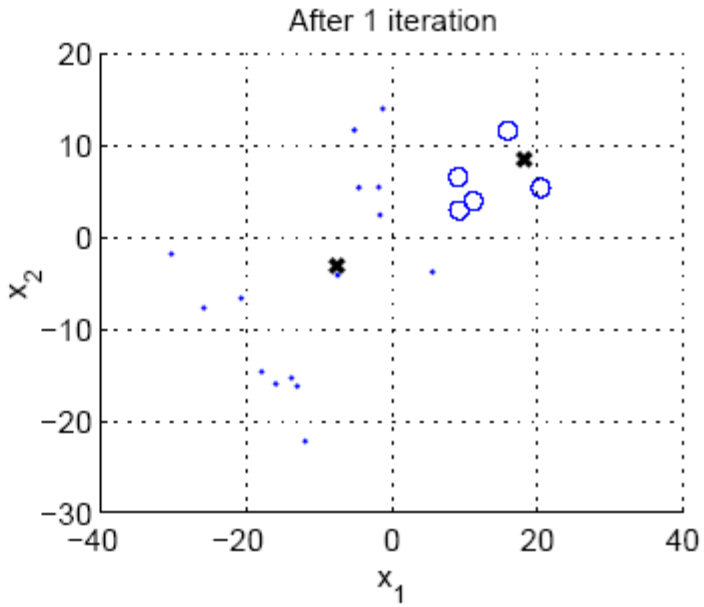
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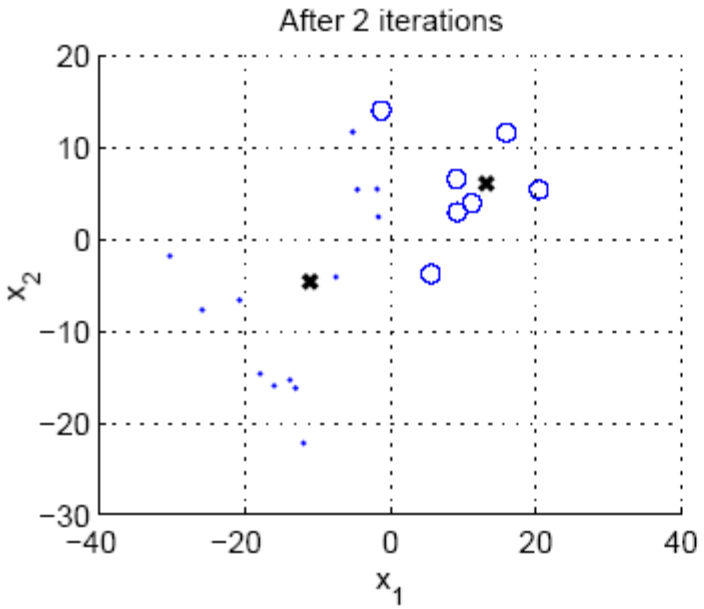
# K-means clustering: other ways to initialize

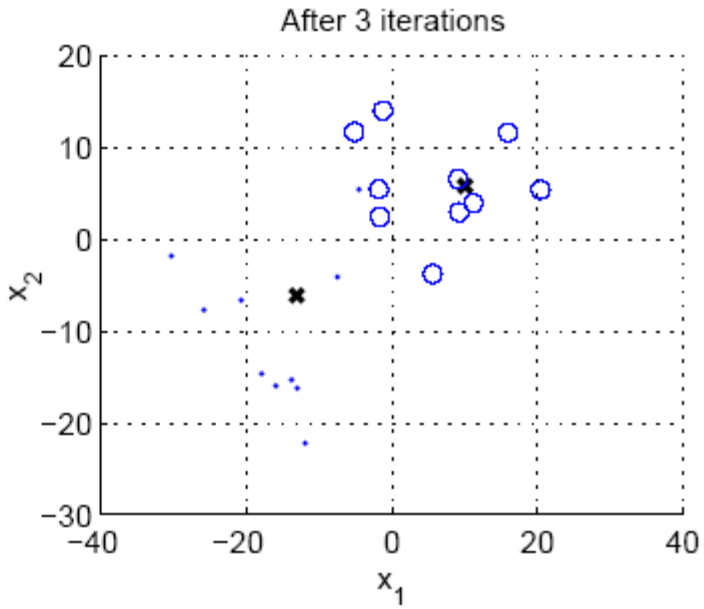



- Use a fraction of data (10%)
- Find centroids
- Use those centroids for the whole data.


k−means: Initial
20
10
0
−10
−20
−30
−40
−20
0
20
40
$x_2$
$x_1$


After 1 iteration
20
10
0
−10
−20
−30
−40
−20
0
20
40
$x_2$
$x_1$


After 2 iterations
20
10
0
−10
−20
−30
−40
−20
0
20
40
$x_2$
$x_1$


After 3 iterations
20
10
0
−10
−20
−30
−40
−20
0
20
40
$x_2$
$x_1$

# K-means clustering: Pros and cons

- Pros
  - Efficient algorithm
  - Widely application to various types of data

- Cons
  - Not suitable for all types of data
  - Initialization problems and outliers
  - restricted to data in which there is a notion of a center

# Mixture Densities

$$p(\mathbf{x}) = \sum_{i=1}^{k} p(\mathbf{x} \mid G_i) P(G_i)$$

where $G_i$ the components/groups/clusters,

$P(G_i)$ mixture proportions (priors),

$p(\boldsymbol{x} \mid G_i)$ component densities

Gaussian mixture where $p(\boldsymbol{x} \mid G_i) \sim N(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$

parameters $\Phi = \{P(G_i), \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i\}_{i=1}^{k}$

unlabeled sample $X=\{\boldsymbol{x}^t\}_t$ (unsupervised learning)

# Clustering using mixture models




- Each distribution corresponds to a cluster

- Find the parameters for each distribution

# Expectation –Maximization (EM)

Select an initial set of model parameters.

**Repeat**

**Expectation Step**

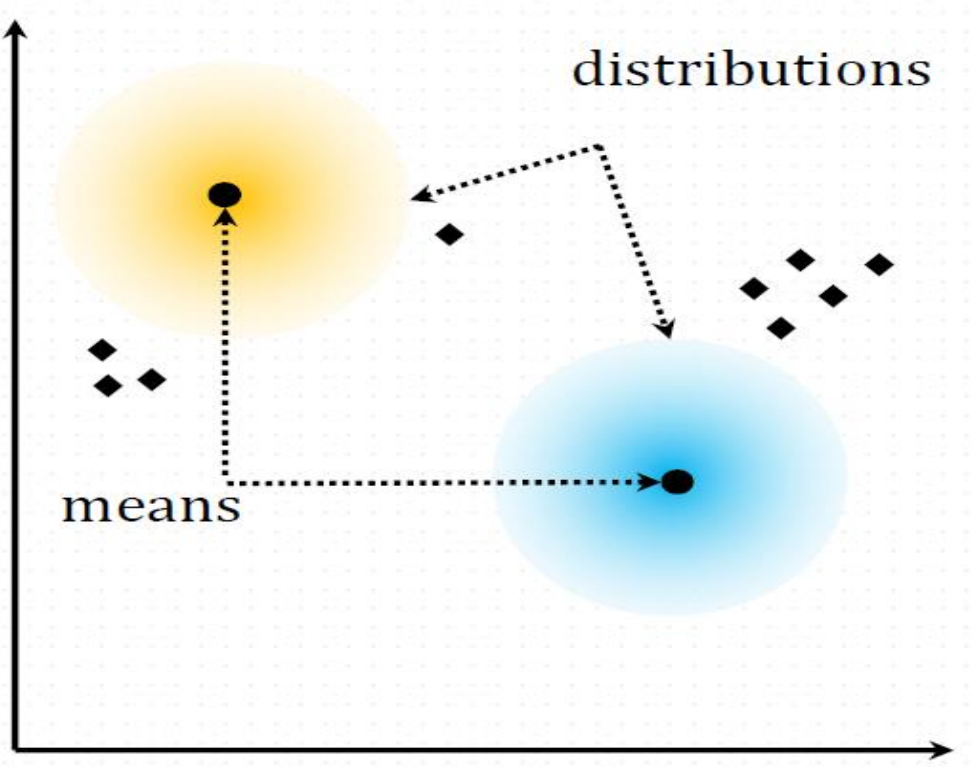
For each object, calculate probability that each object belongs to each distribution,

**Maximization Step**

Given the probabilities from the expectation step, find the new estimates of the parameters that maximize the expected likelihood

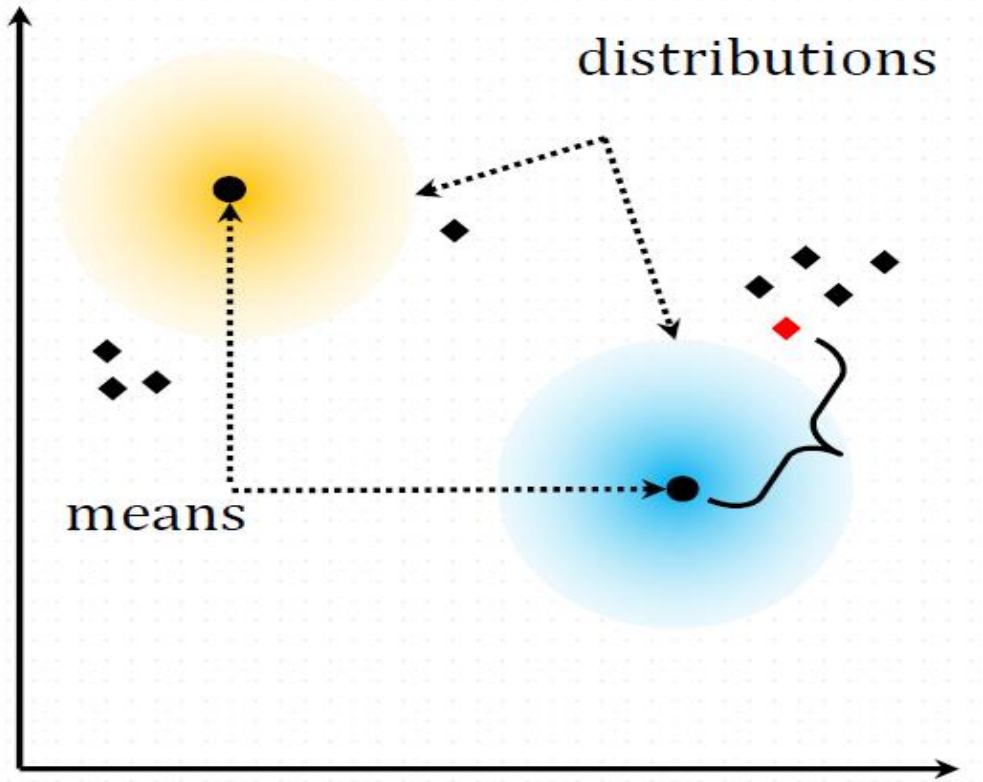
**Until** The parameters do not change (or are below a threshold)

# EM: Visual Idea: Initialize



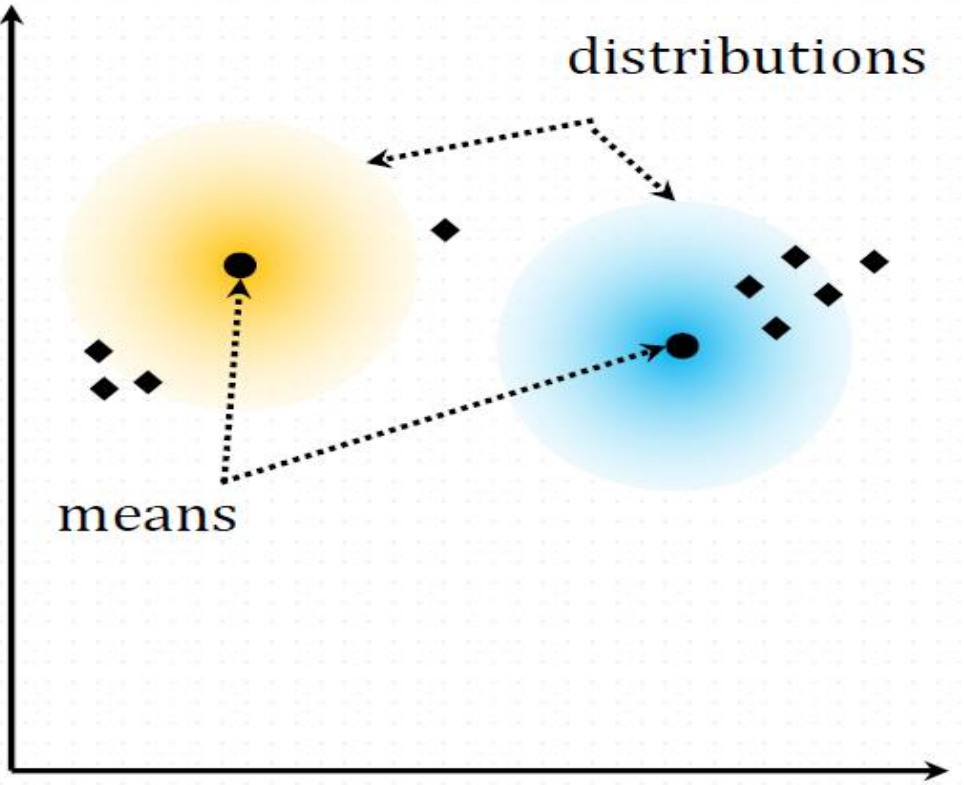

- Make an initial guess for means (initialize varances to some fixed value)

# EM: Visual Idea: Expectation step



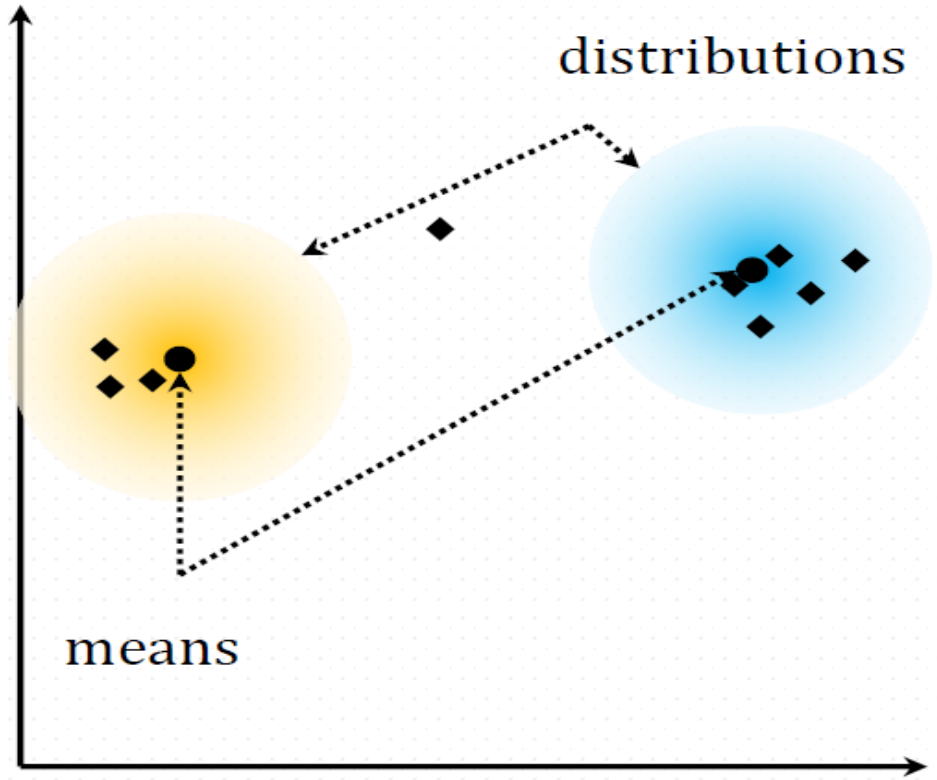

- Compute membership probabilities for each point using bayes rule

# EM: Visual Idea: Maximization step




- Recompute means using weighted (the probabilities that a point belong to a distribution) average of points

# EM: Visual Idea: Finish




- Repeat expectation and maximization step until no or little change (based on threshold)

# Expectation-Maximization (EM)

- Log likelihood with a mixture model

$$\mathcal{L}(\Phi \mid \mathcal{X}) = \log \prod_t p\left(\mathbf{x}^t \mid \Phi\right)$$

$$= \sum_t \log \sum_{i=1}^{k} p\left(\mathbf{x}^t \mid G_i\right) P\left(G_i\right)$$

- Assume hidden variables *z*, which when known, make optimization much simpler
- Complete likelihood, $L_c(\Phi \mid X,Z)$, in terms of ***x*** and ***z***
- Incomplete likelihood, $L(\Phi \mid X)$, in terms of ***x***

# E- and M-steps

**Iterate the two steps**

1. **E-step**: Estimate *z* given X and current Φ
2. **M-step:** Find new Φ' given *z*, X, and old Φ.

$$\text{E-step: } \mathcal{Q}\left(\Phi \mid \Phi^{l}\right) = E\left[\mathcal{L}_c\left(\Phi \mid \mathcal{X}, \mathcal{Z}\right) \mid \mathcal{X}, \Phi^{l}\right]$$

$$\text{M-step: } \Phi^{l+1} = \underset{\Phi}{\operatorname{argmax}}\, \mathcal{Q}\left(\Phi \mid \Phi^{l}\right)$$

An increase in Q increases incomplete likelihood

$$\mathcal{L}\left(\Phi^{l+1} \mid \mathcal{X}\right) \geq \mathcal{L}\left(\Phi^{l} \mid \mathcal{X}\right)$$



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# EM in Gaussian Mixtures

- $z^t_i = 1$ if $\boldsymbol{x}^t$ belongs to $G_i$, 0 otherwise (labels $\boldsymbol{r}^t_i$ of supervised learning); assume $p(\boldsymbol{x}|G_i) \sim N(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$
- E-step:

$$E\left[z_i^t \middle| \mathcal{X}, \Phi^l\right] = \frac{p\left(\mathbf{x}^t | G_i, \Phi^l\right) P\left(G_i\right)}{\sum_j p\left(\mathbf{x}^t | G_j, \Phi^l\right) P\left(G_j\right)}$$

$$= P\left(G_i | \mathbf{x}^t, \Phi^l\right) \equiv h_i^t$$

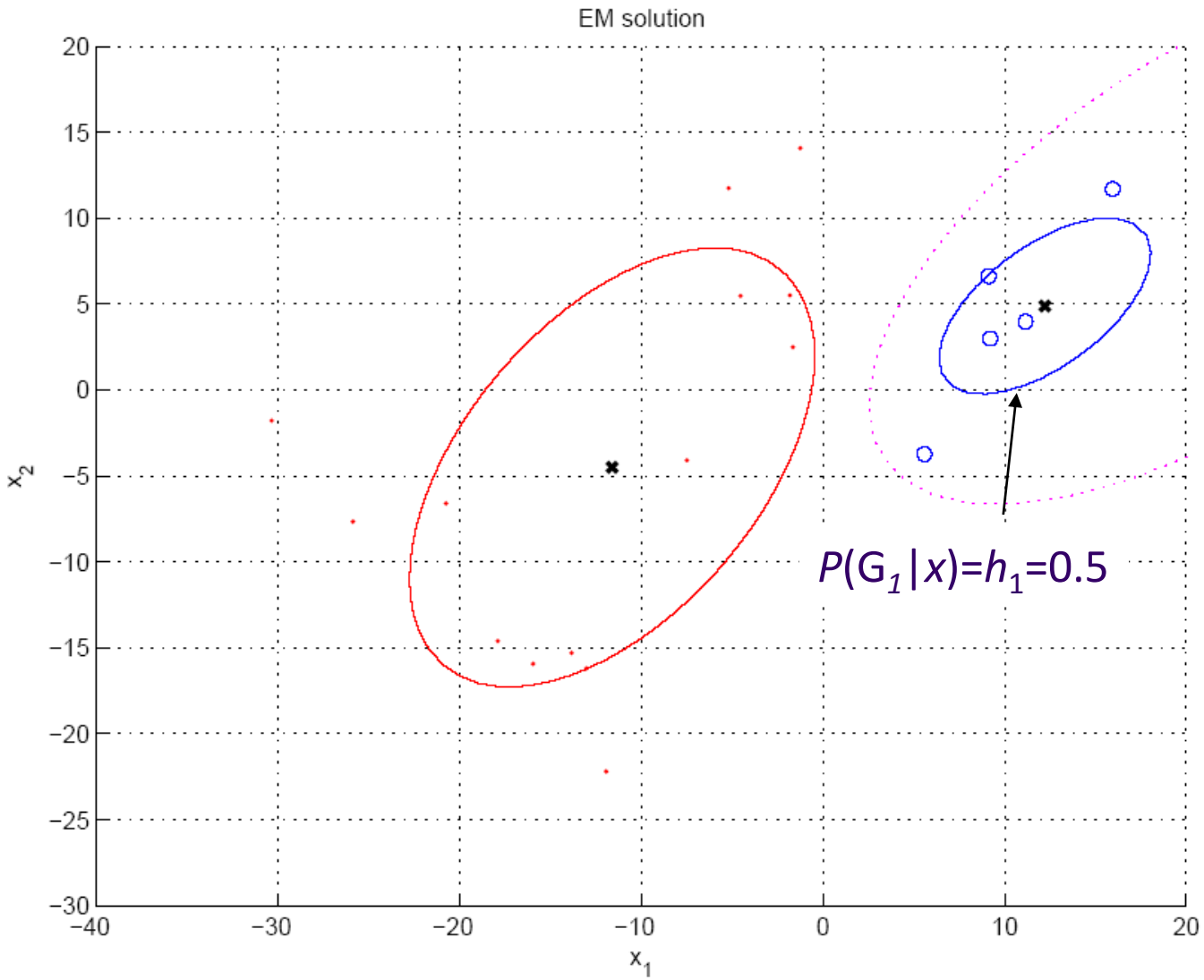
- M-step:

$$P\left(G_i\right) = \frac{\sum_t h_i^t}{N} \qquad \mathbf{m}_i^{l+1} = \frac{\sum_t h_i^t \mathbf{x}^t}{\sum_t h_i^t}$$

$$\mathbf{S}_i^{l+1} = \frac{\sum_t h_i^t \left(\mathbf{x}^t - \mathbf{m}_i^{l+1}\right)\left(\mathbf{x}^t - \mathbf{m}_i^{l+1}\right)^T}{\sum_t h_i^t}$$

*Use estimated labels in place of unknown labels*

# EM : Solution


EM solution
20
15
10
5
0
−5
−10
−15
−20
−25
−30
−40
−30
−20
−10
0
10
20
x2
x1
P(G1|x)=h1=0.5

# Mixture model clustering using EM: Pros and Cons




- Cons
  - Slow
  - Does not work with few data points
  - Does not work with collinear points
  - Challenging to choose number of clusters
  - Challenging to choose distributions

- Pros
  - Can use distribution of various types
  - Easy Characterization of clusters

# Hierarchical Clustering

- Set of nested clusters organized as a tree
- Cluster based on similarities/distances
- Distance measure between instances $\boldsymbol{x}^r$ and $\boldsymbol{x}^s$

Minkowski ($L_p$) (Euclidean for $p = 2$)

$$d_m\left(\mathbf{x}^r, \mathbf{x}^s\right) = \left[\sum_{j=1}^{d}\left(x_j^r - x_j^s\right)^p\right]^{1/p}$$

City-block distance

$$d_{cb}\left(\mathbf{x}^r, \mathbf{x}^s\right) = \sum_{j=1}^{d}\left|x_j^r - x_j^s\right|$$



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# Hierarchical clustering

- Does not assume value of k
- Could be more meaningful than K-Mean.

# Agglomerative vs Decisive clustering

- Agglomerative (Bottom up):
  - Start with N group, merge them based on similarity

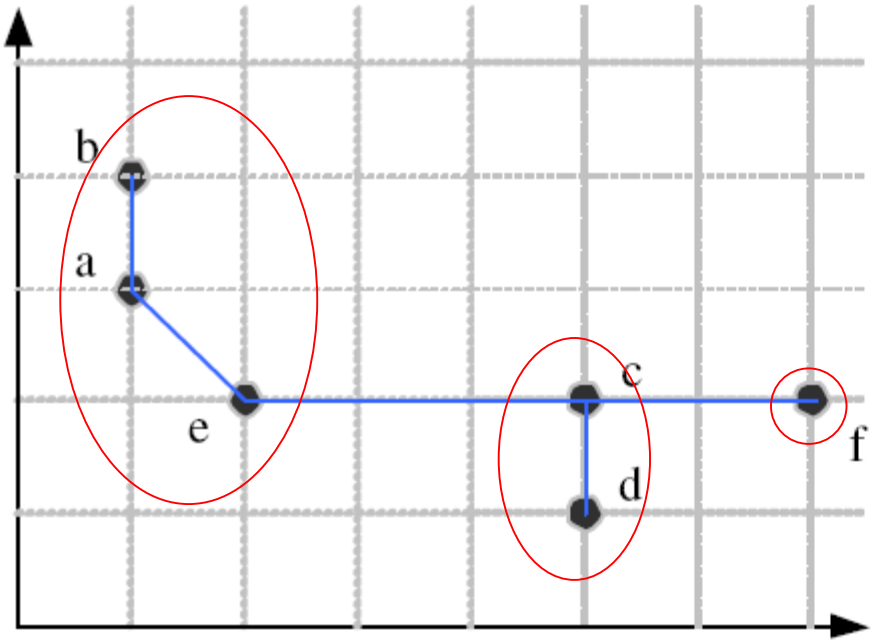
- Decisive (Top Down):
  - Start with one group, divide them into smaller group until each group has a single instance.

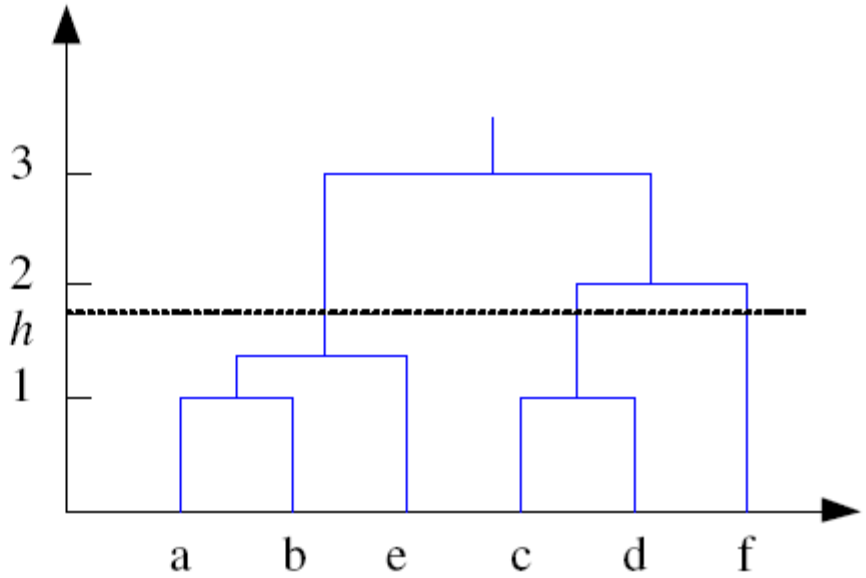
# Agglomerative Clustering: details

- Start with *N* groups each with one instance and merge two closest groups at each iteration

- Distance between two groups $G_i$ and $G_j$:
  - Single-link:
$$d(G_i, G_j) = \min_{\mathbf{x}^r \in G_i, \mathbf{x}^s \in G_j} d(\mathbf{x}^r, \mathbf{x}^s)$$
  - Complete-link:
$$d(G_i, G_j) = \max_{\mathbf{x}^r \in G_i, \mathbf{x}^s \in G_j} d(\mathbf{x}^r, \mathbf{x}^s)$$
  - Average-link: average of distances between all pairs
  - Centroid: distance between the centroids

# Example: Single-Link Clustering




*Dendrogram*

# Evaluation of clustering algorithm




- Unsupervised
  - Without any external information.
- Supervised
  - compare with external structure.
- Relative
  - Compares different clusterings or

# Cluster Validity

$$overall\ validity = \sum_{i=1}^{K} w_i\ validity(C_i)$$

- *Validity*($C_i$)
  - Cohesion: sum of the proximities with respect to prototype (mean/median etc.)
  - Separation: relative proximity of clusters

# The Silhouette Coefficient

- For i[th] object $s_i = (b_i - a_i)/\max(a_i, b_i)$
  - $a_i$: average distance to all other objects in its cluster
  - $b_i$: minimum of object's average distances to all the objects in a different cluster
- Combines cohesion and separation
- Possible value can be between -1 and 1
- A positive value is desired with $a_i < b_i$ and $a_i$ close to 0.

# How to Choose *k*

- Defined by the application
- Plot data (after PCA) and check for clusters
- Incremental (leader-cluster) algorithm: Add one at a time until “elbow” (reconstruction error/log likelihood/intergroup distances)
- Manually check for meaning
- Rule of thumb k = sqrt(n/2), not very reliable